Marcos Martinez Jaime

Full Stack & Systems Developer

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Github · Portfolio · LinkedIn
Buenos Aires, Argentina

> Developer specializing in **high-concurrency web applications, multi-tenant architectures, and low-level data structures**. Architect of a production multi-tenant SaaS serving manufacturing enterprises daily. Bridges distributed backends (PostgreSQL RLS, Redis atomic locks, Supabase Realtime) with systems programming in **C++20 & Rust** and **UI/UX Design Systems in Figma and modern React.**

Professional Experience

Full Stack & Systems Developer
01/2025 – Present

Industrial Operations SaaS Platform — In Daily Production

- Architected and deployed an enterprise **multi-tenant SaaS across 14 modules** in active daily production for manufacturing companies, replacing disconnected spreadsheets with centralized budgeting, planning, and supply chain.
- Engineered a **2D cutting stock optimization algorithm** for panel nesting on standard sheet materials, calculating exact edge-banding meters and minimizing raw material waste before workshop dispatch.
- Implemented database security with **PostgreSQL Row-Level Security (RLS)** via Prisma and granular RBAC, restricting workshop tablets exclusively to cut and assembly queues while safeguarding company financial margins.
- Designed real-time floor dispatch using **Supabase Realtime broadcast channels**, synchronizing work orders, cut lists, and inventory movements with zero database polling overhead.

`TypeScript` `React 19` `Next.js` `Node.js` `Prisma ORM` `PostgreSQL (RLS)` `Supabase Realtime` `Tailwind CSS`

Mobile & UI/UX Developer Intern
03/2024 – 11/2024

SmartBuy Mobile Application

- Co-developed a collaborative mobile solution eliminating supermarket checkout lines by generating a virtual shopping cart through real-time barcode scanning and instant in-app digital checkout.
- Implemented mobile UI and state management in **Flutter & Dart**, integrated camera scanning pipelines, and designed interactive product prototypes and user journeys in **Figma**.

`Flutter` `Dart` `Figma UX/UI` `Design Systems` `Prototyping`

Systems & Core Engineering Projects

Concurrent LRU Cache
C++20 · CMake · GoogleTest

Modern C++20 Thread-Safe In-Memory Cache

- Implemented a thread-safe in-memory cache achieving amortized **O(1) time complexity** for lookup and eviction via a doubly linked list and hash map topology.
- Synchronized concurrent access using **std::shared_mutex** (reader-writer lock) for high parallel throughput; validated with GoogleTest.

LLM Inference Engine
Rust · GGUF · Transformer

From-Scratch High-Performance Language Model Runtime in Pure Rust

- Architected a from-scratch tensor engine and decoder-only Transformer in **pure Rust (zero ML frameworks / no PyTorch)**, supporting GGUF model parsing via mmap, RoPE positional embeddings, GQA attention, and RMSNorm.
- Implemented an **autoregressive KV cache with zero-allocation decoding** on CPU and INT8/INT4 quantization (Q8_0, Q4_0).

Education & Certifications

Systems Analyst (In Progress)
08/2025 – Present

Universidad Da Vinci

IT Technician (7-Year Technical High School Degree)
2018 – 2024

Instituto Técnico Industrial San Judas Tadeo

Comprehensive 7-year technical education covering relational database administration (SQL Server, MySQL, Oracle, DDL/DML), low-level systems programming, Linux/Windows administration, networks, and agile methodologies.

Artificial Intelligence Specialization
07/2025 – 11/2025

Samsung Innovation Campus

Technical Expertise & Languages

Languages & Core:	Rust, C++20, TypeScript, JavaScript, SQL, Dart, HTML5, CSS3, Data Structures & Algorithms.
Backend & Systems:	Node.js, Next.js (App Router), Express, Prisma ORM, PostgreSQL (RLS), Supabase, Redis, REST APIs.
AI & Automation:	AI Agent Workflows, Process Automation, LLM API Integration (Tool Use, Prompt Engineering).
UI/UX & Product Design:	Figma (Design Systems, Auto-Layout, Tokens, Prototyping), React 19, Tailwind CSS v4, Motion, WCAG.
DevOps & Tools:	Git, GitHub, Docker, CI/CD, CMake, Linux, GoogleTest, Catch2.
Languages:	**English: C1 · Spanish: Native.**